Application Deployment

(Deploy the given react application to a production ready state)

Application:
Clone the below mentioned repo and deploy the application. (Run the application in port 80 [HTTP])
Repo URL: https://github.com/sriram-R-krishnan/devops-build

```
DevOps_Project_3_Reactjs_E-commerce_Application/
├── Dockerfile
├── Jenkinsfile
├── README.md
├── ansible
│   ├── inventory.ini
│   ├── sg-rules.yml
│   └── vars.yml
├── build.sh
├── deploy.sh
├── devops-build
│   └── build
│       ├── _redirects
│       ├── asset-manifest.json
│       ├── favicon.ico
│       ├── index.html
│       ├── logo192.png
│       ├── logo512.png
│       ├── manifest.json
│       ├── robots.txt
│       └── static
│           ├── css
│           │   ├── main.cf5c13c5.css
│           │   └── main.cf5c13c5.css.map
│           └── js
│               ├── 787.2f5360e2.chunk.js
│               ├── 787.2f5360e2.chunk.js.map
│               ├── main.f1c48542.js
│               ├── main.f1c48542.js.LICENSE.txt
│               └── main.f1c48542.js.map
├── docker-compose.yml
└── terraform
    └── main.tf

8 directories, 25 files
```

Docker:

- Dockerize the application by creating a Dockerfile

```
FileName: Dockerfile

FROM nginx:alpine

# Remove default nginx content
RUN rm -rf /usr/share/nginx/html/*

# Copy static build
COPY devops-build/build /usr/share/nginx/html

# Expose port 80
EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

- Create a docker-compose file to use the above image

```
version: '3'
services:
  react-app:
    image: ghostatdocker/react-app-prod:latest
    container_name: react-app
    ports:
      - "80:80"
    restart: always
~
```

Bash Scripting:
Write 2 scripts

- build.sh – for building docker images

```
#!/bin/bash

BRANCH=$1

if [ "$BRANCH" == "prod" ]; then
    docker build -t ghostatdocker/react-app-prod:latest .
else
    docker build -t ghostatdocker/react-app-dev:latest .
fi
~
```

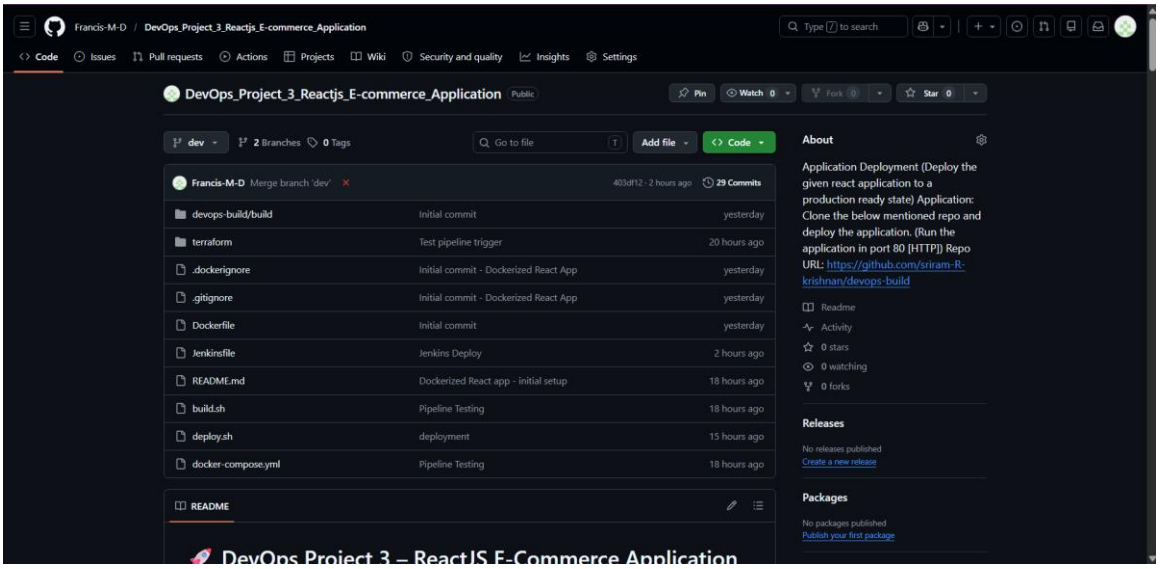
- deploy.sh – for deploying the image to server

```
#!/bin/bash

docker-compose down --remove-orphans
docker rm -f react-app 2>/dev/null || true
docker-compose pull
docker-compose up -d --force-recreate
~
```

Version Control:

- Push the code to github to dev branch (use dockerignore & gitignore files)

```
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git remote -v
origin  https://github.com/Francis-M-D/DevOps_Project_3_Reactjs_E-commerce_Application.git (fetch)
origin  https://github.com/Francis-M-D/DevOps_Project_3_Reactjs_E-commerce_Application.git (push)
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git branch
* dev
  master
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git status
On branch dev
Your branch is up to date with 'origin/dev'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        ansible/

nothing added to commit but untracked files present (use "git add" to track)
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$
```

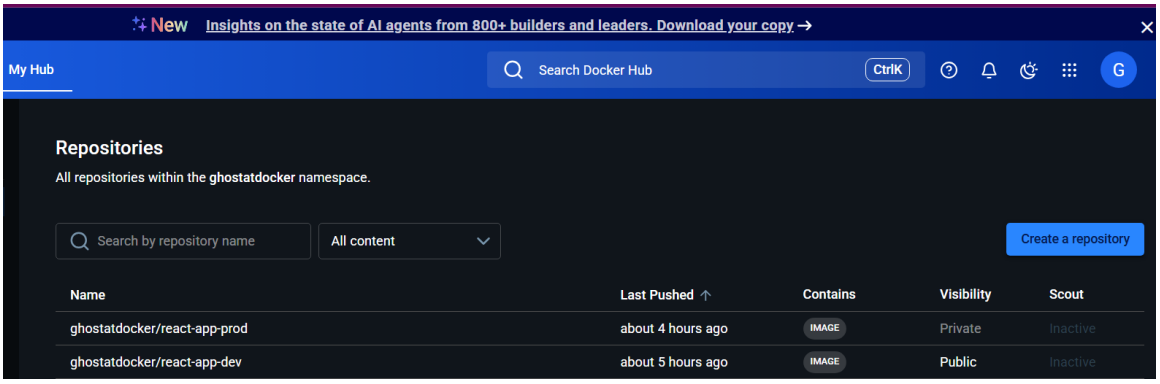


- Note: Use only CLI for related git commands

```
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git add .
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git status
On branch dev
Your branch is up to date with 'origin/dev'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   ansible/inventory.ini
        new file:   ansible/sg-rules.yml
        new file:   ansible/vars.yml

GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git commit -m "Adding ansible"
[dev 65be44a] Adding ansible
 3 files changed, 61 insertions(+)
 create mode 100644 ansible/inventory.ini
 create mode 100644 ansible/sg-rules.yml
 create mode 100644 ansible/vars.yml
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 1.01 KiB | 517.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Francis-M-D/DevOps_Project_3_Reactjs_E-commerce_Application.git
   403df12..65be44a  dev -> dev
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$
```

Docker hub:

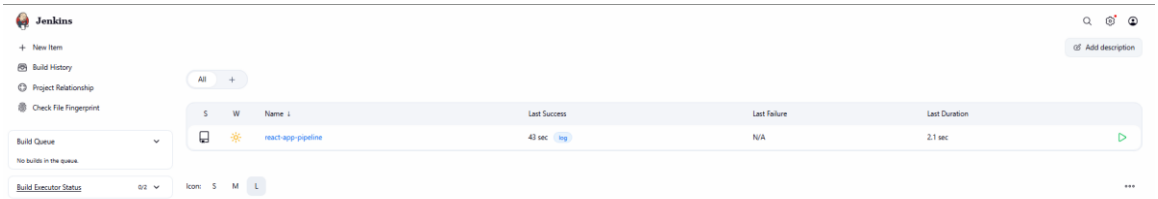
- Create 2 repos “dev” and “prod” to push images
- “Prod” repo must be private and “dev” repo can be public



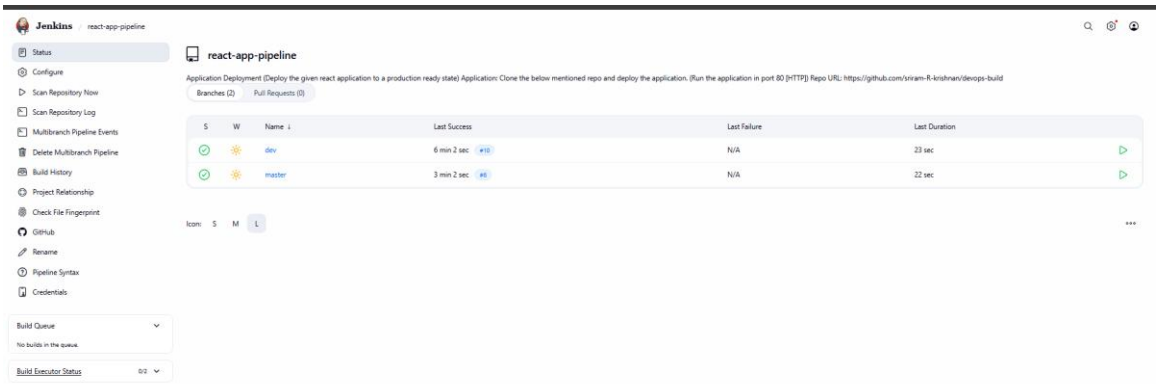
Jenkins:

- Install and configure jenkins build step as per needs to build, push & deploy the application

```
ubuntu@ip-172-31-43-40:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/lib/systemd/system/jenkins.service; enabled; vendor preset: enabled)
     Active: active (running) since Sun 2026-04-19 21:16:54 UTC; 42min ago
   Main PID: 17713 (java)
      Tasks: 84 (limit: 2266)
     Memory: 584.8M
        CPU: 1min 59.732s
     CGroup: /system.slice/jenkins.service
             └─17713 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Apr 19 21:50:09 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:50:09.011+0000 [id=688]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
Apr 19 21:51:09 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:51:09.022+0000 [id=698]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
Apr 19 21:52:09 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:52:09.005+0000 [id=705]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
Apr 19 21:53:09 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:53:09.023+0000 [id=714]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
Apr 19 21:54:09 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:54:09.042+0000 [id=725]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
Apr 19 21:55:09 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:55:09.057+0000 [id=735]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
Apr 19 21:56:08 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:56:08.980+0000 [id=829]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
Apr 19 21:57:09 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:57:09.058+0000 [id=836]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
Apr 19 21:58:09 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:58:09.067+0000 [id=845]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
Apr 19 21:59:09 ip-172-31-43-40 jenkins[17713]: 2026-04-19 21:59:09.008+0000 [id=942]        INFO        o.j.p.g.ApiRateLimitChecker$RateLimitCheckerAdapte>
lines 1-20/20 (END)
```



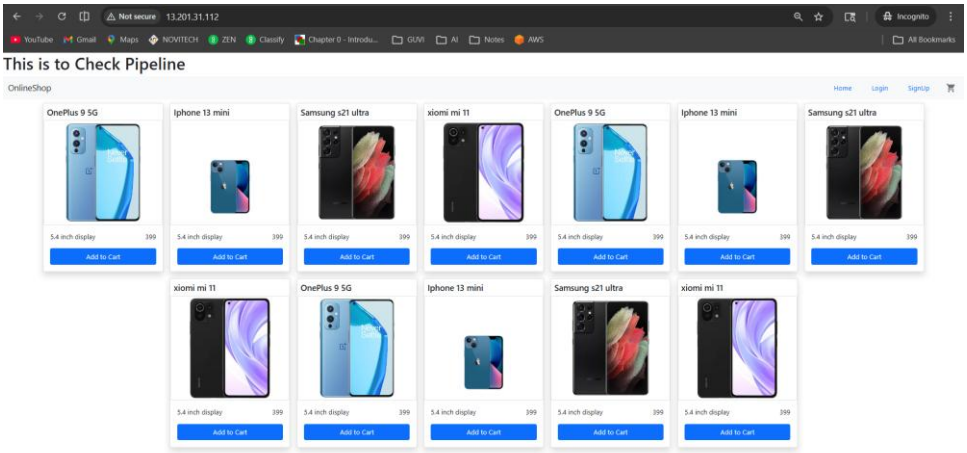
- Connect jenkins to the github repo with auto build trigger from both dev & master branch



- If code pushed to dev branch, docker image must build and pushed to dev repo in docker hub

```
        stage('Deploy') {
            when {
                expression {
                    return env.BRANCH_NAME == 'master'
                }
            }
            steps {
                sh 'bash deploy.sh'
            }
        }
```

[Note: “This is to check Pipeline” is showing, I am going to remove that and push the code to dev]



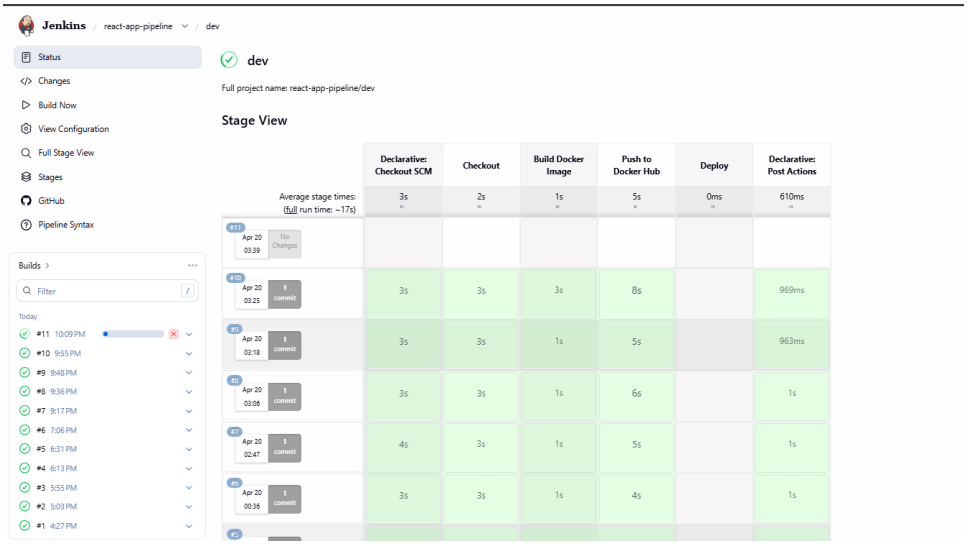
Changing the index.html file:

```
GHOST@GHOST: ~/Final_Proj   ubuntu@ip-172-31-43-40: ~
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ cat devops-build/build/index.html
<!doctype html>
<html lang="en">
        <head><meta charset="utf-8"/>
                <link rel="icon" href="/favicon.ico"/>
                <meta name="viewport" content="width=device-width,initial-scale=1"/>
                <meta name="theme-color" content="#000000"/>
                <meta name="description" content="Web site created using create-react-app"/>
                <link rel="apple-touch-icon" href="/logo192.png"/>
                <link rel="manifest" href="/manifest.json"/>
                <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-1BmE4kWBq78iYhFldvKuhfTAU6auU8tT94
WrHftjDbrCEXSU1oBoqyl2QvZ6jIW3" crossorigin="anonymous">
                <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.5/font/bootstrap-icons.css">
                <title>React App</title>
                <script defer="defer" src="/static/js/main.f1c48542.js"></script>
                <link href="/static/css/main.cf5c13c5.css" rel="stylesheet">
        </head>
        <body>
                <noscript>You need to enable JavaScript to run this app.</noscript>
                <h1>This is to Check Pipeline</h1>
                <div id="root"></div>
        </body>
</html>
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ vi devops-build/build/index.html
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ cat devops-build/build/index.html
<!doctype html>
<html lang="en">
        <head><meta charset="utf-8"/>
                <link rel="icon" href="/favicon.ico"/>
                <meta name="viewport" content="width=device-width,initial-scale=1"/>
                <meta name="theme-color" content="#000000"/>
                <meta name="description" content="Web site created using create-react-app"/>
                <link rel="apple-touch-icon" href="/logo192.png"/>
                <link rel="manifest" href="/manifest.json"/>
                <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.1.3/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-1BmE4kWBq78iYhFldvKuhfTAU6auU8tT94
WrHftjDbrCEXSU1oBoqyl2QvZ6jIW3" crossorigin="anonymous">
                <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.10.5/font/bootstrap-icons.css">
                <title>React App</title>
                <script defer="defer" src="/static/js/main.f1c48542.js"></script>
                <link href="/static/css/main.cf5c13c5.css" rel="stylesheet">
        </head>
        <body>
                <noscript>You need to enable JavaScript to run this app.</noscript>
                <div id="root"></div>
        </body>
</html>
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$
```

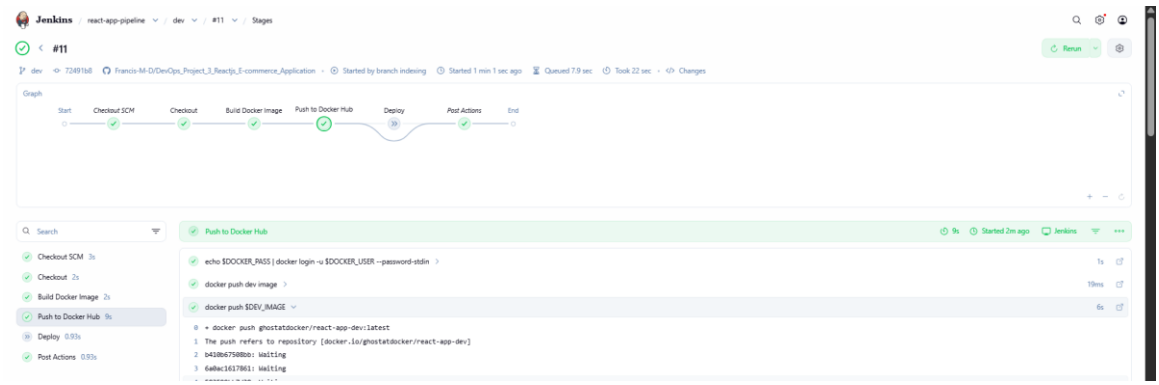
Pushing it to dev repo:

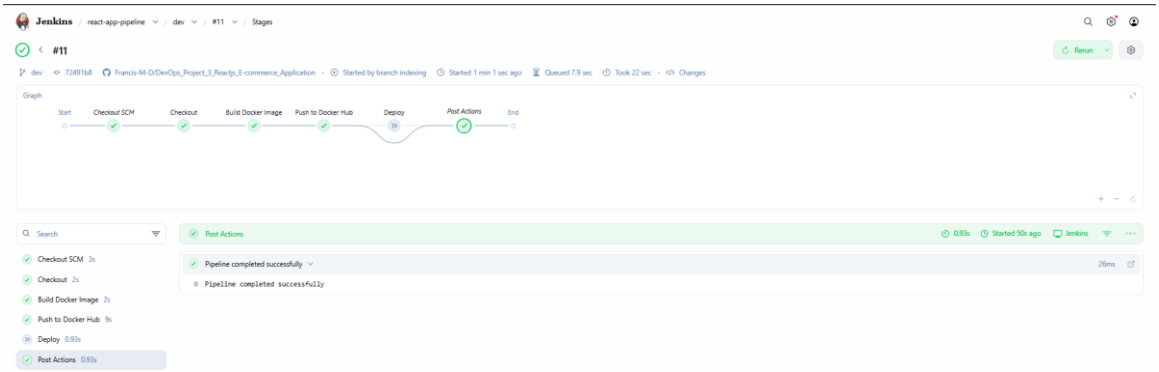
```
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git branch
* dev
  master
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git status
On branch dev
Your branch is up to date with 'origin/dev'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   devops-build/build/index.html

no changes added to commit (use "git add" and/or "git commit -a")
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git add .
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git commit -m "Dev Repo Trigger"
[dev 72491b8] Dev Repo Trigger
 1 file changed, 1 deletion(-)
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 400 bytes | 200.00 KiB/s, done.
Total 5 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/Francis-M-D/DevOps_Project_3_Reactjs_E-commerce_Application.git
   c0e02c9..72491b8  dev -> dev
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$
```

Checking the pipeline triggering due to webhook:

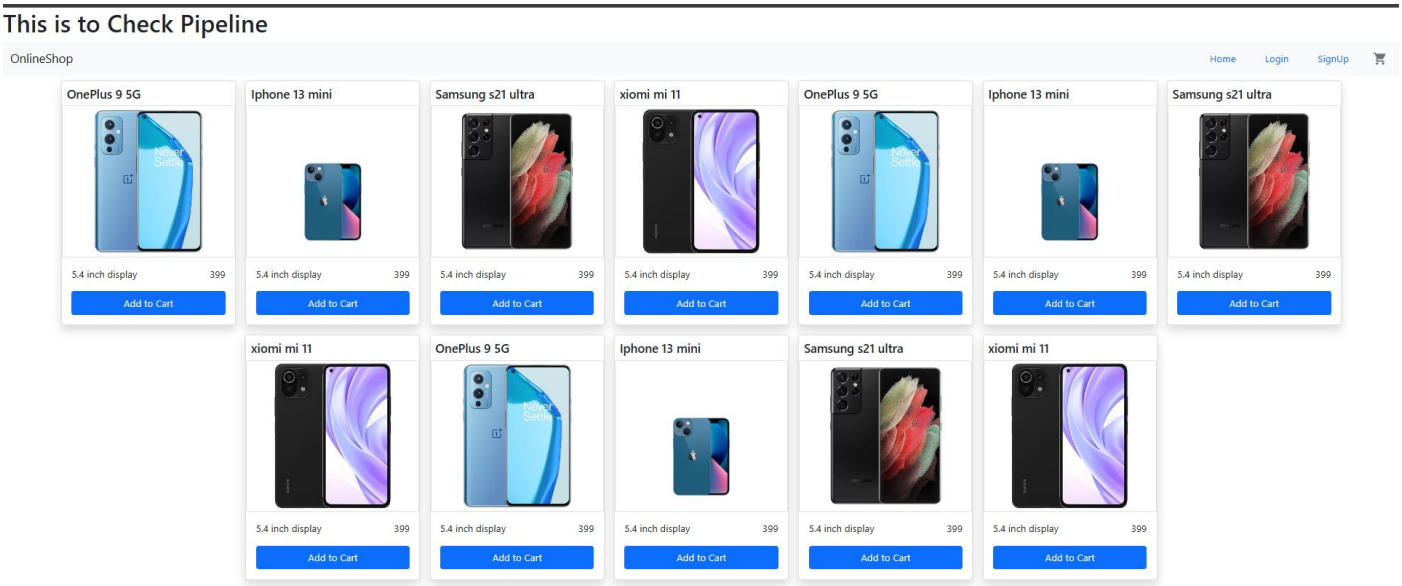


As pushed to dev repo, the image has been built and pushed to docker dev repo:





Checking the application: [No change should have happened]



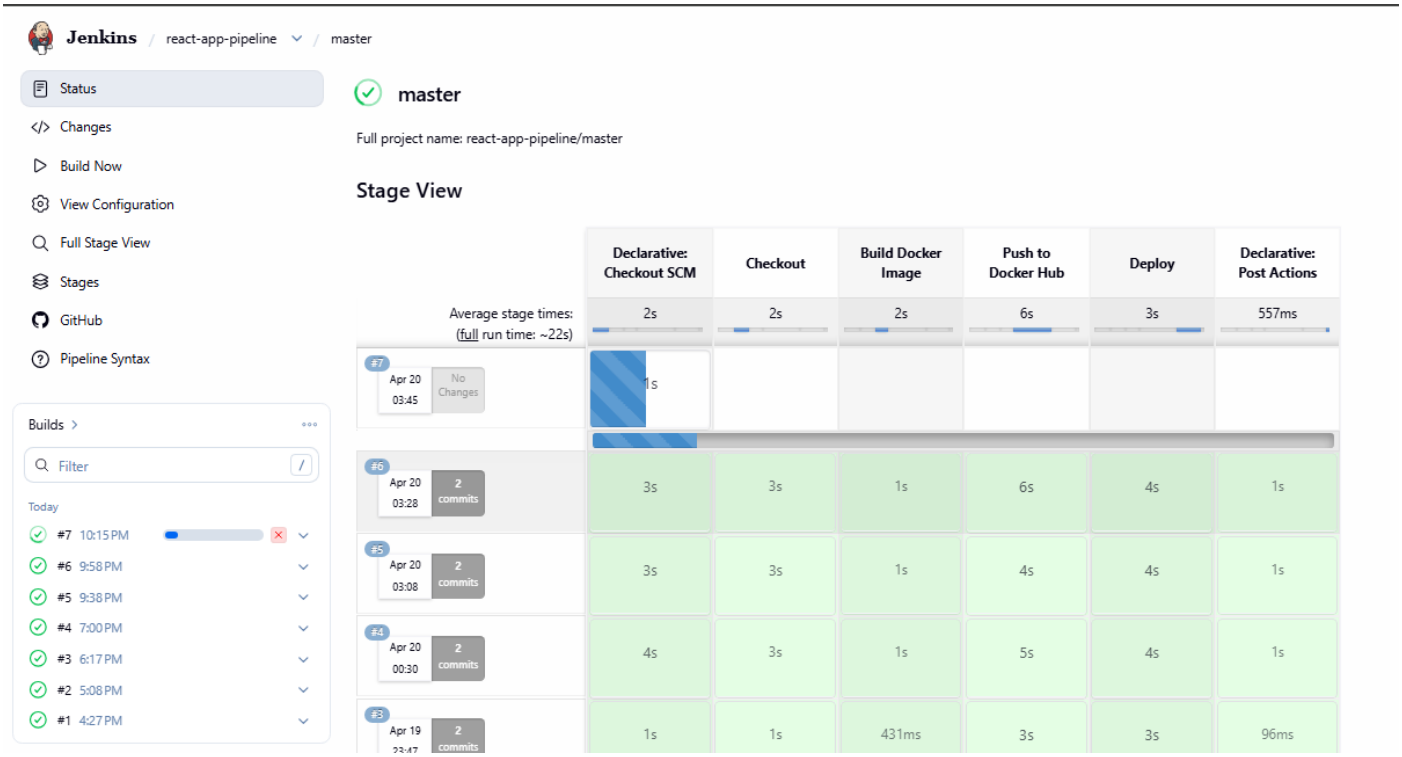
- **If dev merged to master, then docker image must be pushed to prod repo in docker hub**

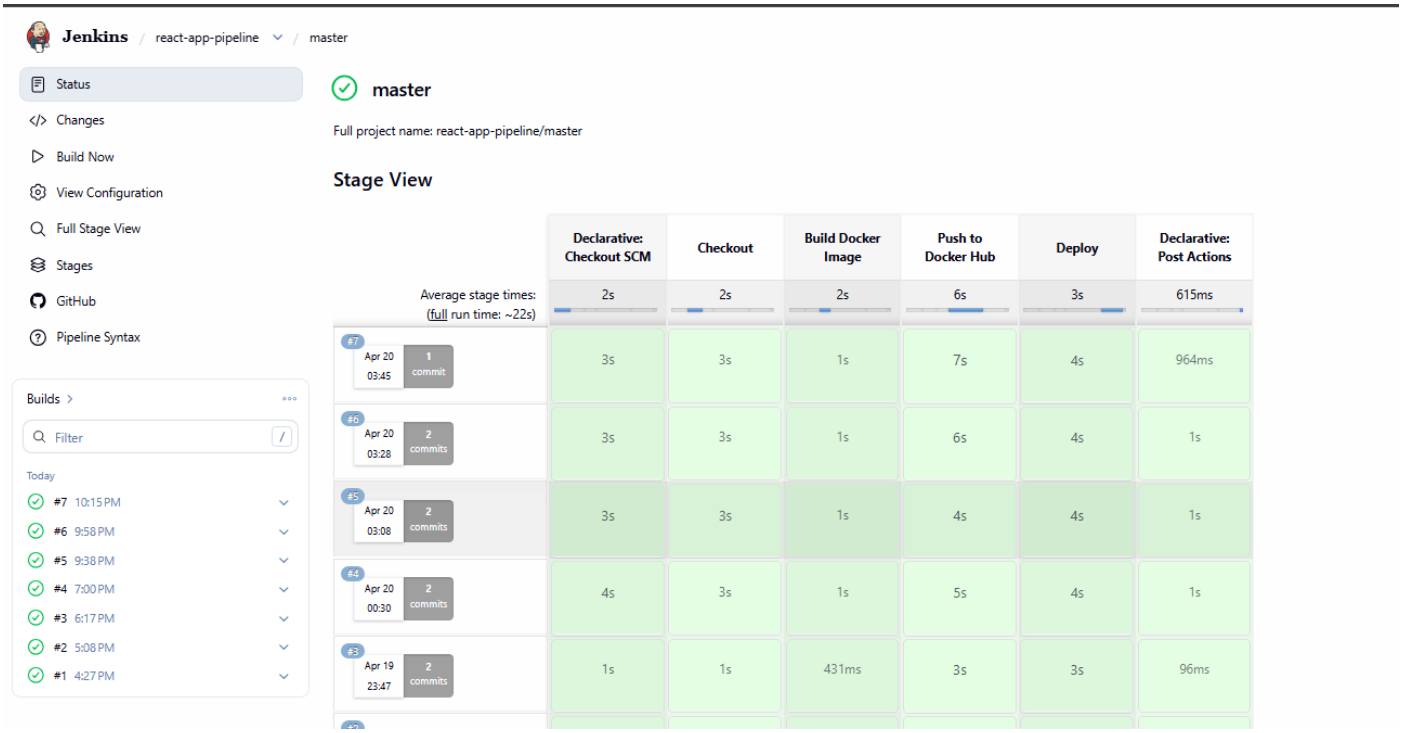
Now I am going to merge the dev repo with master and push it to github:

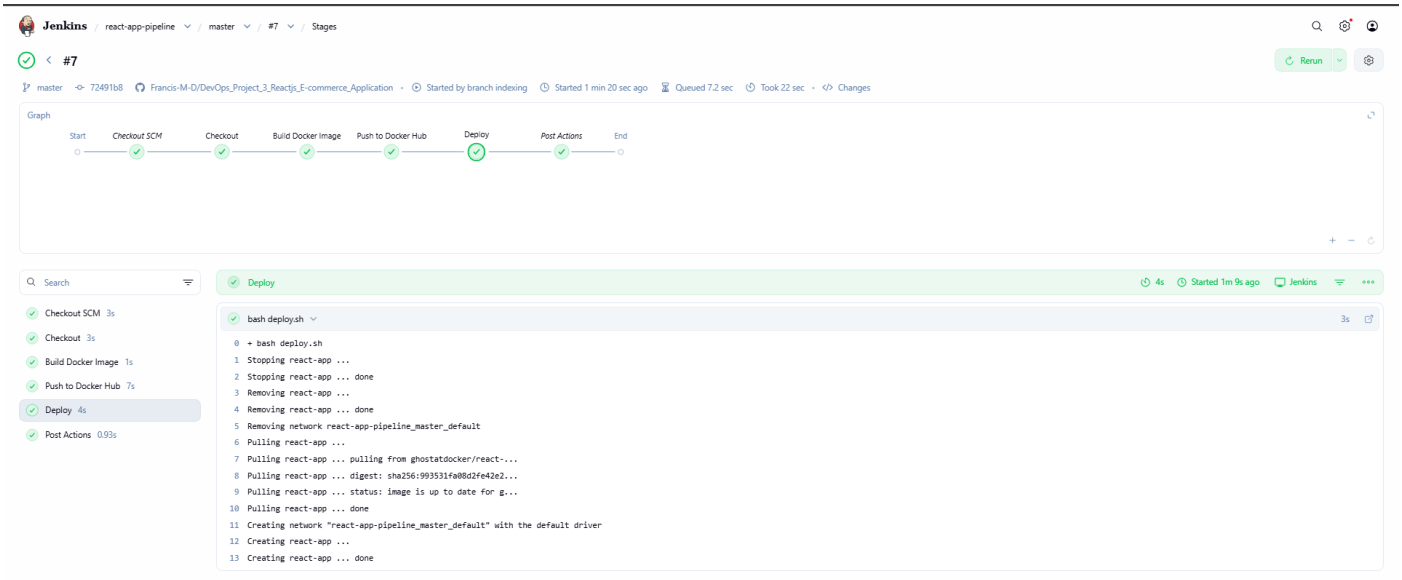
```
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git checkout master
Switched to branch 'master'
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git branch
  dev
* master
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git merge dev
Updating c0e02c9..72491b8
Fast-forward
 devops-build/build/index.html | 1 -
 1 file changed, 1 deletion(-)
```

```
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$ git push origin master
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Francis-M-D/DevOps_Project_3_Reactjs_E-commerce_Application.git
   c0e02c9..72491b8  master -> master
GHOST@GHOST:~/Final_Projects/DevOps_Project_3_Reactjs_E-commerce_Application$
```

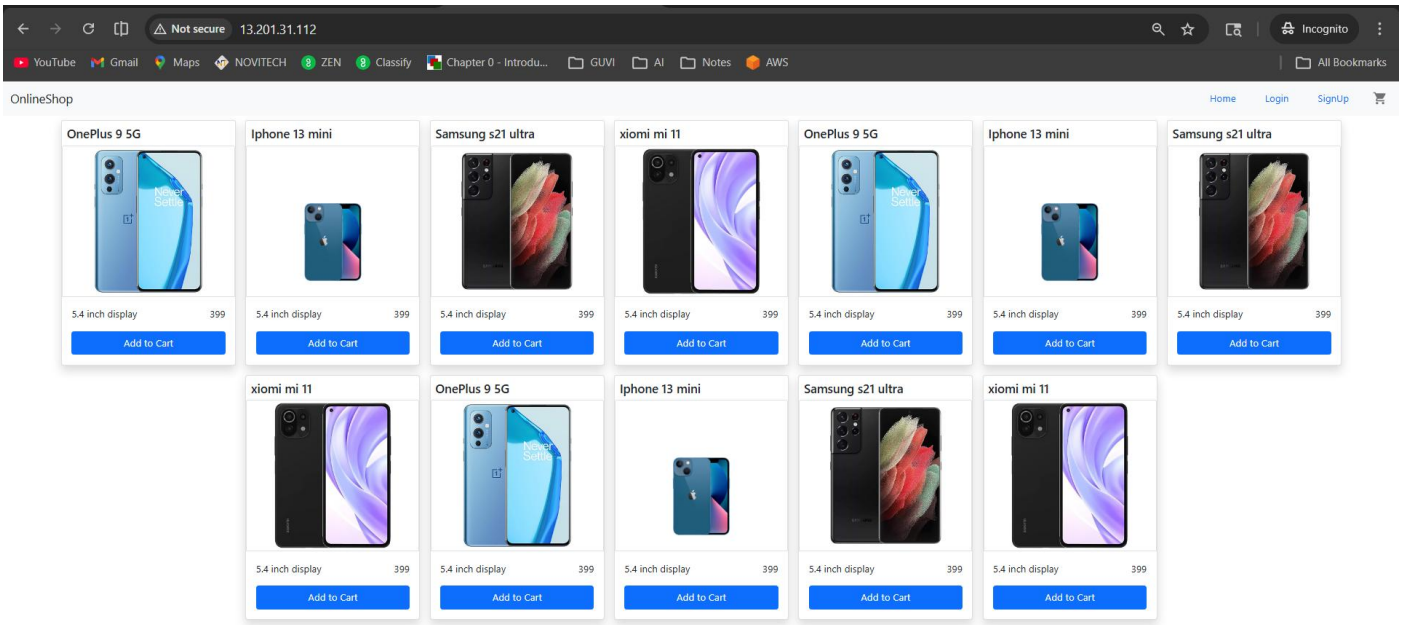
Jenkins Pipeline Triggered from git push:







Verifying the changes in the application:



AWS:
Launch t2.micro instance and deploy the create application.
Configure SG as below:

- Whoever has the ip address can access the application
- Login to server can should be made only from your ip address

[We don't have access to t3.micro instance, hence I am using t3 series instance]

I am using terraform [Left] to create this instance, so I am adding the codes to let everyone access the application.

Using an ansible file [Right], to change the security group to change my IP whenever I want, as I am having dynamic IP in my router.

```
# ----------------------------
# Security Group
# ----------------------------
resource "aws_security_group" "jenkins_sg" {
  name        = "jenkins-sg"
  description = "Allow SSH, Jenkins, HTTP, HTTPS"

  ingress {
    description = "SSH"
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = [var.my_ip]
  }

  ingress {
    description = "Jenkins"
    from_port   = 8080
    to_port     = 8080
    protocol    = "tcp"
    cidr_blocks = [var.my_ip]
  }

  ingress {
    description = "HTTP"
    from_port   = 80
    to_port     = 80
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  ingress {
    description = "HTTPS"
    from_port   = 443
    to_port     = 443
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }

  egress {
    description = "Allow all outbound"
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
```

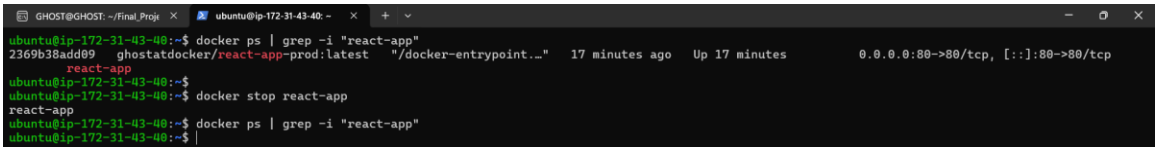
```
---
- name: Configure AWS Security Group Ingress Rules
  hosts: local
  gather_facts: false

  vars_files:
    - vars.yml

  tasks:
    - name: Ensure boto3 is installed
      pip:
        name: boto3
        state: present
        extra_args: "--break-system-packages"

    - name: Add HTTP rule - open to everyone
      amazon.aws.ec2_security_group:
        name: "{{ sg_name }}"
        description: "Security group for DevOps Project 3"
        region: "{{ aws_region }}"
        rules:
          - proto: tcp
            from_port: 80
            to_port: 80
            cidr_ip: "0.0.0.0/0"
            rule_desc: "HTTP - open to everyone"

          - proto: tcp
            from_port: 22
            to_port: 22
            cidr_ip: "{{ my_ip }}"
            rule_desc: "SSH - my IP only"

          - proto: tcp
            from_port: 8080
            to_port: 8080
            cidr_ip: "{{ my_ip }}"
            rule_desc: "Jenkins - my IP only"

          - proto: tcp
            from_port: 3001
            to_port: 3001
            cidr_ip: "{{ my_ip }}"
            rule_desc: "Uptime Kuma - my IP only"

        rules_egress:
          - proto: all
"ansible/sg-rules.yml" 56L, 1422B
```

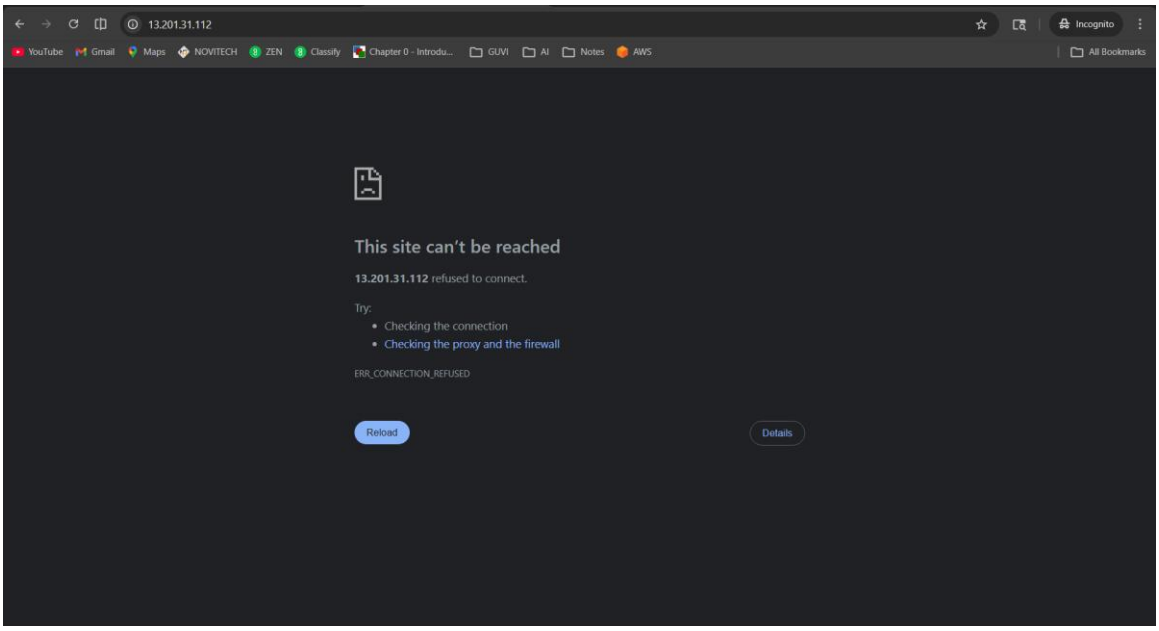
Monitoring:

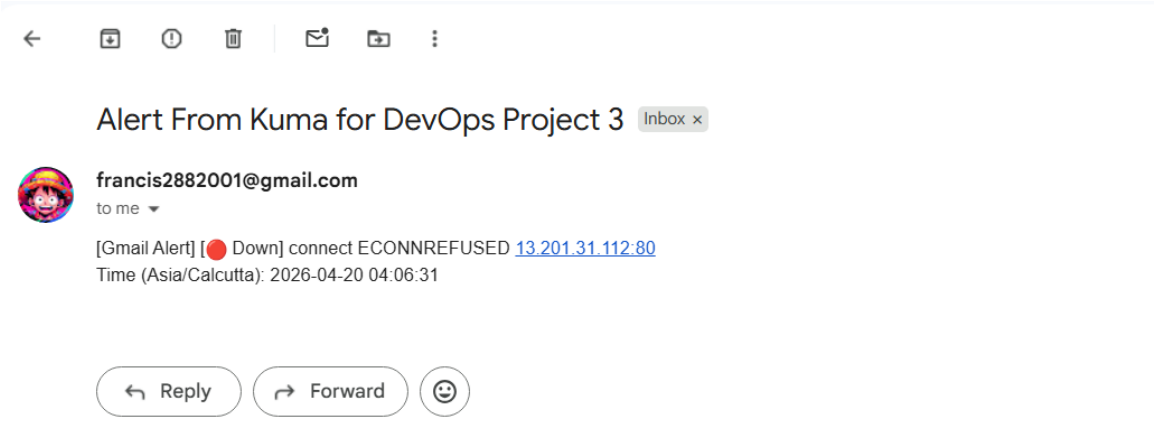
- Setup a monitoring system to check the health status of the application. (Open-source)

- Sending notifications only if the application goes down is highly appreciable

[Going to stop the application to trigger the alert]



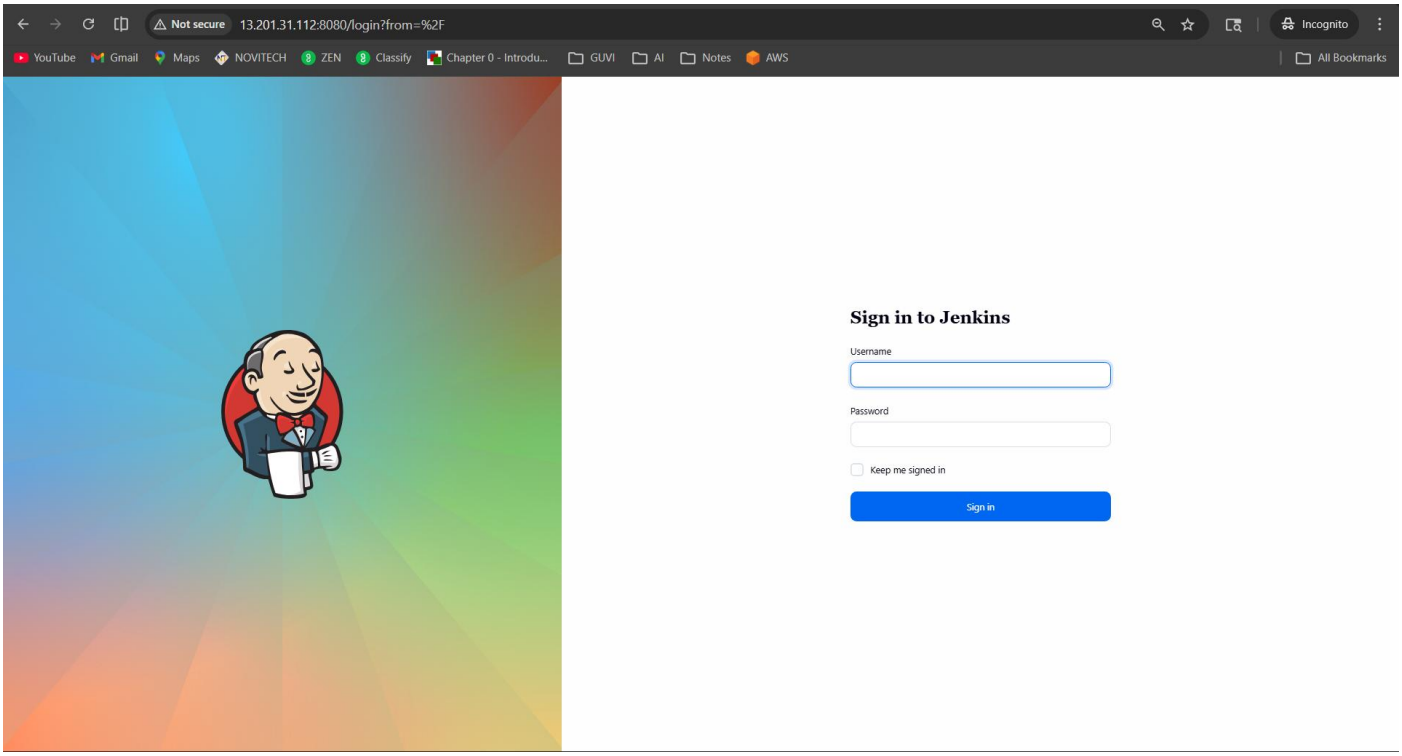




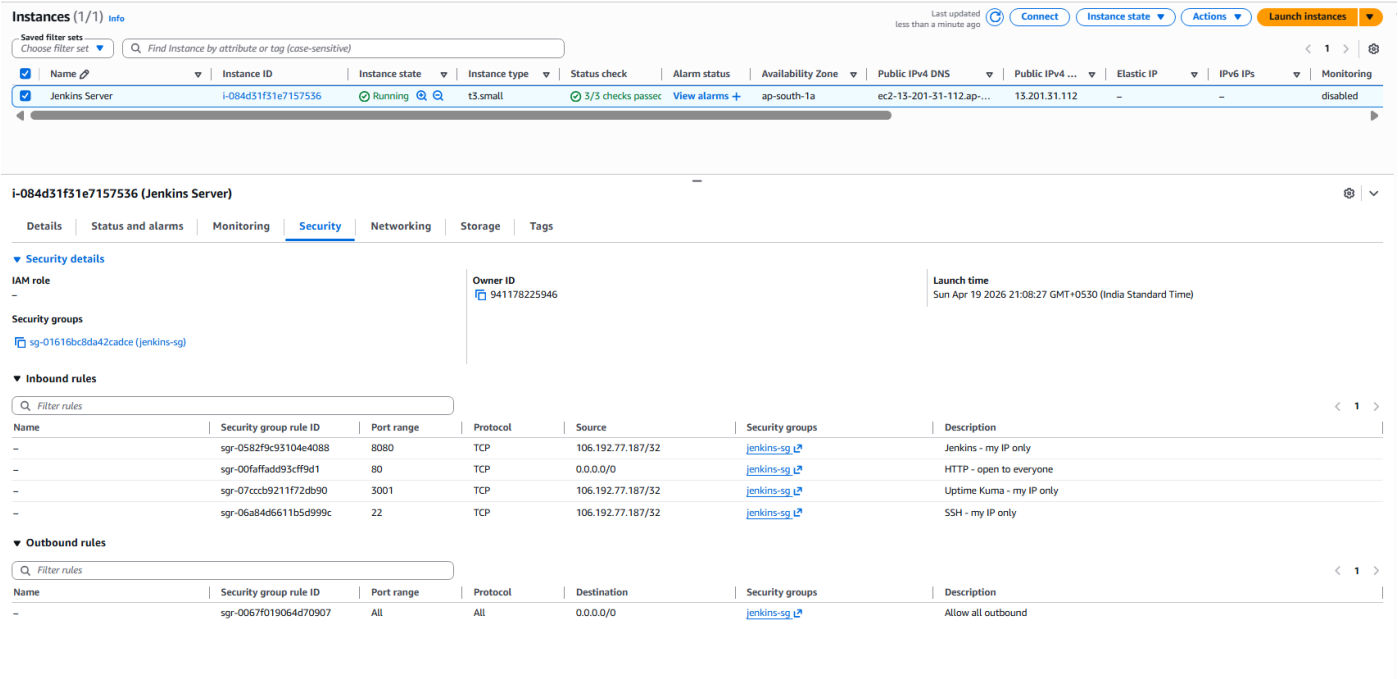
[Received mail alert, when the container goes down]

Submission:

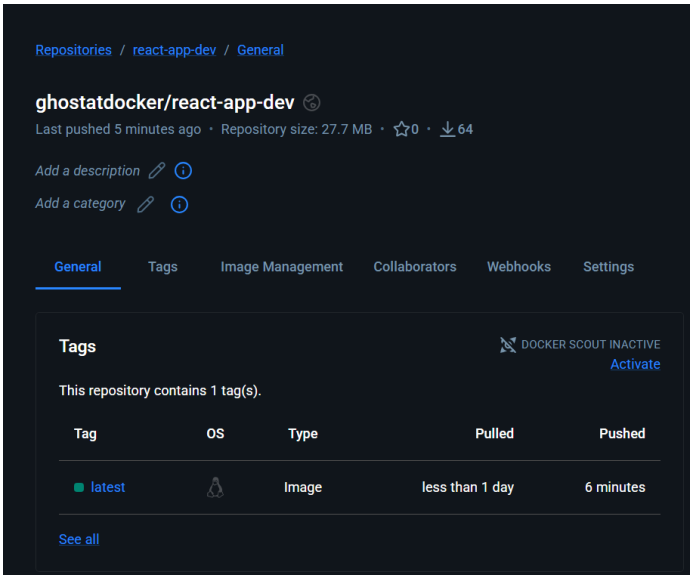
- Github repo URL, deployed site URL, docker images name must be added in the submission
  - https://github.com/Francis-M-D/DevOps_Project_3_Reactjs_E-commerce_Application.git
  - http://13.201.31.112/
  - ghostatdocker/react-app-dev
  - ghostatdocker/react-app-prod
- Upload the screenshots of below mentioned to github repo:
  - Jenkins (login page, configuration settings, execute step commands)

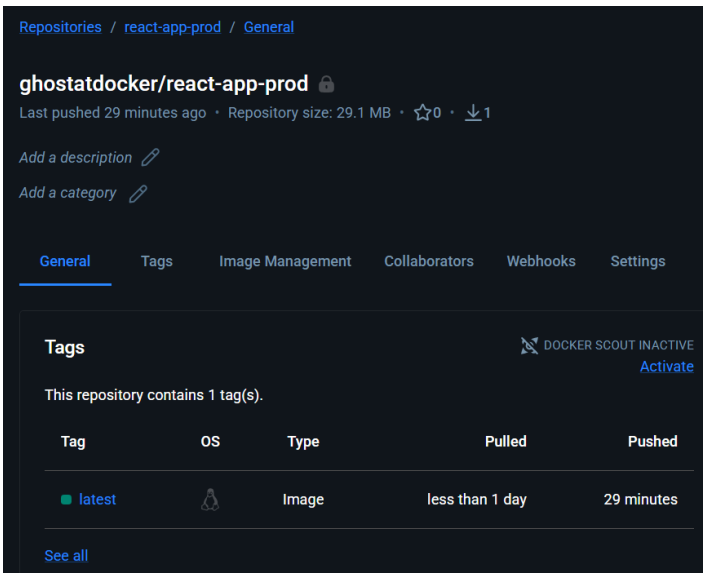


  - AWS (EC2-Console, SG configs)

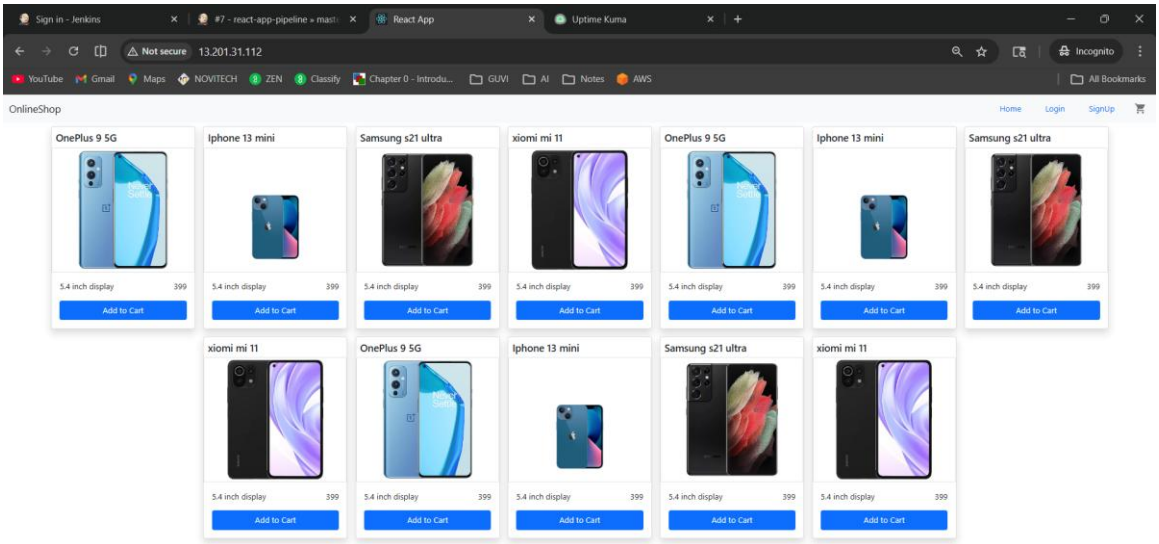
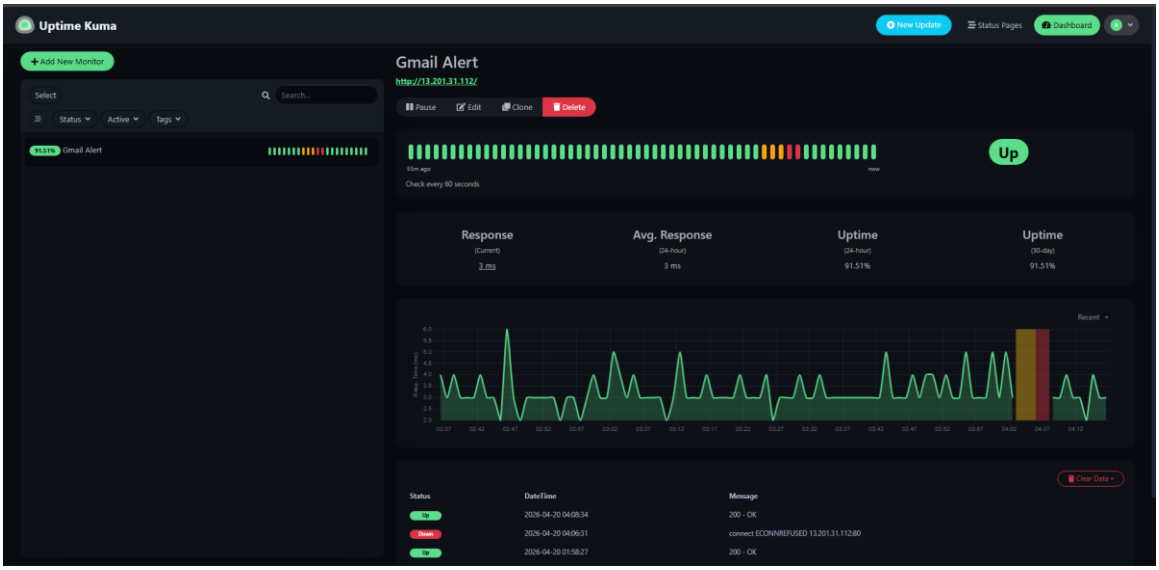


- Docker hub repo with image tags





- Deployed site page



- Monitoring health check status